Lecture 20
Confidence Levels and t Intervals

Review

1 **In "we are 95% confident", what is the 95% about?**

The method: about 95% of intervals built this way capture the parameter.

2 **What is the 95% interval for p, and when does it work?**

$\hat{p} \pm 2\sqrt{(\hat{p}(1-\hat{p}) / n)}$, when $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$.

3 **Where does the 2 in "± 2 SE" come from?**

95% of a normal curve lies within 1.96 SD of the mean; 2 is 1.96 rounded.

Why 95%?

Every interval on Monday was estimate ± 2 SE, which captures the parameter about 95% of the time.

A cardiologist wants to be 99% sure instead.

(a) Should the interval get wider or narrower?
(b) What in "estimate ± 2 SE" has to change?

(a) Wider: more confidence needs a bigger net
(b) The 2 becomes a bigger number, the critical value

Confidence levels

confidence level C

the share of all possible samples whose interval captures the parameter: 90%, 95%, 99%

α

the share that miss: $\alpha = 1 - C$, split as $\alpha/2$ in each tail

critical value $z_{1-\alpha/2}$

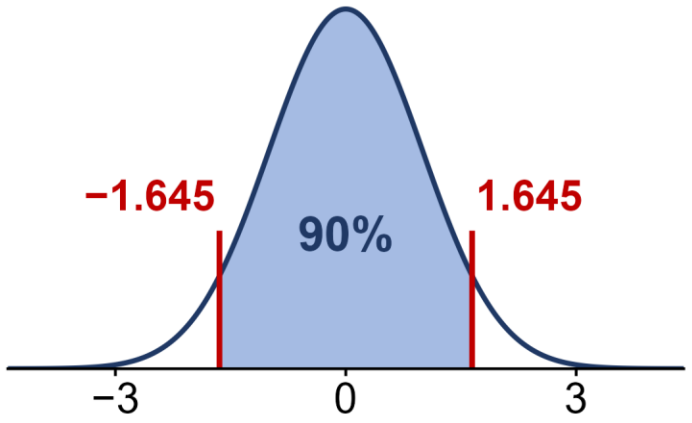
the z with area $1 - \alpha/2$ to its left; the interval is estimate $\pm$ z $\times$ SE

Where z comes from

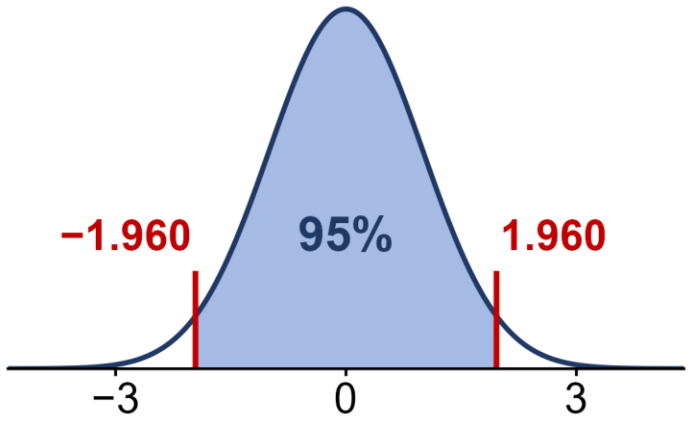
The middle C of the standard normal curve is shaded; $\alpha/2$ is left in each tail.

For a 90% interval, what area to the LEFT of z do we look up in the z-table?

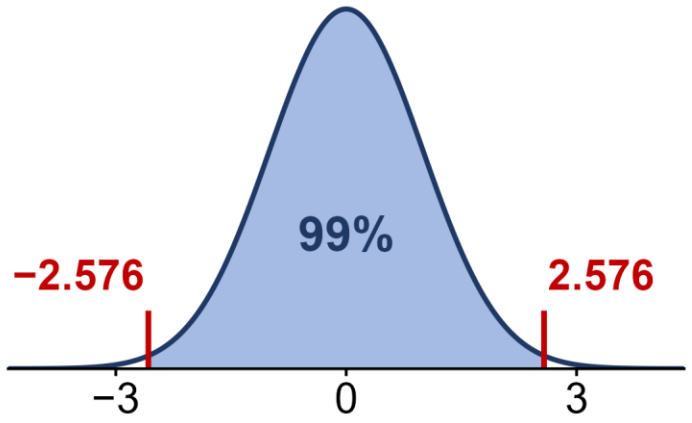
90%: 0.05 in each tail



95%: 0.025 in each tail



99%: 0.005 in each tail



90%: 1 − 0.05 = 0.95 → z = 1.645 **95%: 0.975 → 1.960** **99%: 0.995 → 2.576**

Example: critical values

Find the area $1 - \alpha/2$ inside the z-table, then read off z.

C = 80%; C = 90%; C = 99%

(a) What is $\alpha/2$ for each?
(b) What area to the left do we look up?
(c) What is the critical value?

80%: $\alpha/2 = 0.10$, area 0.90 → z = 1.28
90%: $\alpha/2 = 0.05$, area 0.95 → z = 1.645 (halfway between 1.64 and 1.65)
99%: $\alpha/2 = 0.005$, area 0.995 → z = 2.576 (between 2.57 and 2.58)

Data: a gene and a drug

Clopidogrel (Plavix), a blood thinner used after heart attacks and stents, is switched on by the liver enzyme CYP2C19. People with the CYP2C19*2 variant switch on less of it.

X = 1 if a person carries at least one copy of *2, and 0 otherwise

PERSON	POPULATION	GENOTYPE	COPIES OF *2	X
HG00403	Southern Han Chinese	G/G	0	0
HG00404	Southern Han Chinese	A/A	2	1
HG00406	Southern Han Chinese	G/A	1	1
HG00407	Southern Han Chinese	G/G	0	0
HG00409	Southern Han Chinese	G/G	0	0
…	…	…	…	…

n = 504 East Asian participants of the 1000 Genomes Project, which sequenced 2,504 people worldwide to catalog human genetic variation

Example: carriers of CYP2C19*2

Estimate p, the proportion of all individuals of East Asian descent who carry the *2 variant.

277 of n = 504 people are carriers

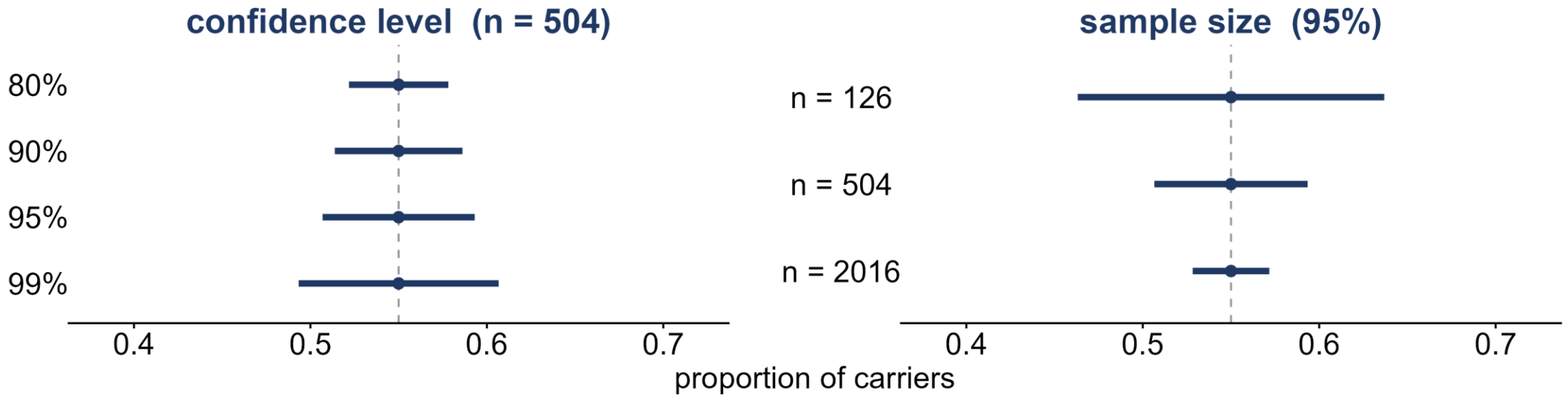
(a) What is $\hat{p}$? Is the check met? What is the standard error?
(b) What are the 90%, 95% and 99% confidence intervals for p?
(c) Can we say more than half of people carry *2?

(a) $\hat{p} = 277 / 504 \approx 0.550$; 277 and 227 are both > 5
SE = $\sqrt{(0.550 \times 0.450 / 504)} \approx 0.022$
(b) 90%: 0.550 ± 1.645(0.022): 0.514 to 0.586
95%: 0.550 ± 1.960(0.022): 0.507 to 0.593
99%: 0.550 ± 2.576(0.022): 0.493 to 0.607
(c) At 90% and 95%, yes; the 99% interval includes 0.5

Making an interval narrower

The carrier interval ($\hat{p}$ = 0.550) at four confidence levels, and at 95% with 4 times fewer or more people.

What happens to the width as C goes up? As n goes up?



Higher C → wider. 4 times the n → half the width (SE has √n on the bottom)

What if σ is unknown?

Every interval for μ so far used σ, the population standard deviation.

$\bar{x} \pm z\, \sigma / \sqrt{n}$

(a) If we don't know μ, do we know σ?

(b) What could we use in its place?

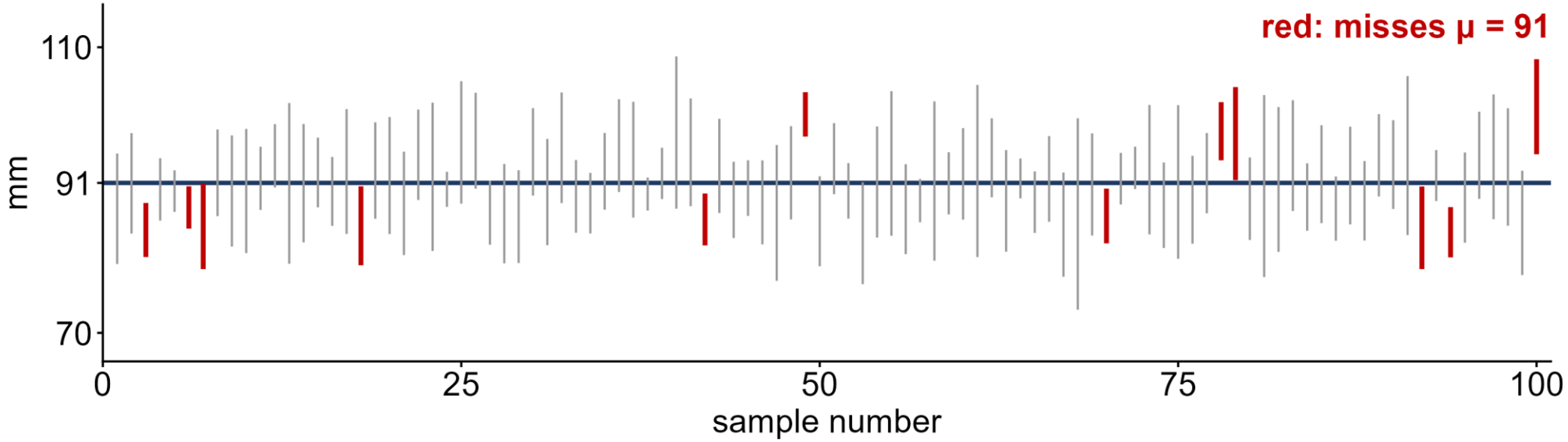
(a) Almost never

(b) s, the sample standard deviation: $\bar{x} \pm z\, s / \sqrt{n}$. Does that still work?

Does s in place of σ work?

Simulated: 100 samples of n = 5 whiskers from N(91, 8) mm; each line is $\bar{x} \pm 1.96\ s / \sqrt{5}$.

A 95% method should miss about 5 times. How many of these 100 miss μ?



12 of 100: s changes from sample to sample, so the intervals are too narrow too often

The t distribution

t distribution

a bell curve like N(0, 1) but with heavier tails; use it when s replaces σ

degrees of freedom

df = n − 1; the more df, the closer t is to N(0, 1)

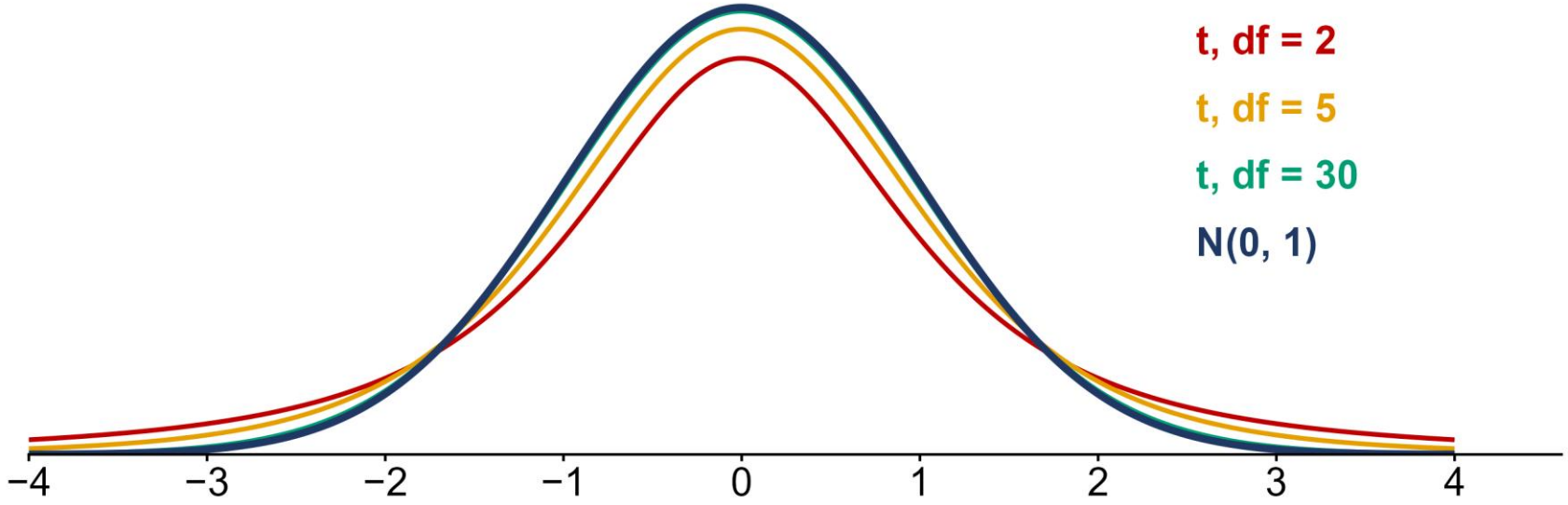
t critical value $t_{n-1,\,1-\alpha/2}$

from the t-table row for df = n − 1; larger than z, so the interval is wider

t versus the standard normal

Density curves of t with 2, 5 and 30 degrees of freedom, and the standard normal N(0, 1).

Which curve has the heaviest tails? What happens as df grows?



df = 2. As df grows, t approaches N(0, 1); by df = 30 they nearly match

Example: reading the t-table

In the course t-table, the row is df = n − 1 and the column is the confidence level.

95%, n = 10; 90%, n = 25; 99%, n = 10

(a) What is the t critical value for each?

(b) How does each compare with z for the same confidence level?

(a) df = 9: 2.262 df = 24: 1.711 df = 9: 3.250

(b) z = 1.960, 1.645, 2.576: t is always larger

Which interval?

a proportion p

$\hat{p} \pm z \sqrt{\hat{p}(1 - \hat{p}) / n}$, when $n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$

a mean μ, σ known

$\bar{x} \pm z \sigma / \sqrt{n}$

a mean μ, σ unknown

$\bar{x} \pm t s / \sqrt{n}$, with $df = n - 1$

when the t interval works

the population is roughly normal, or $n \geq 30$

Data: seal whiskers

Ginter et al. (2012) measured the longest undamaged whisker of 92 seals and sea lions from 11 species, to study how whisker shape evolved.

X = the total length of a seal's longest whisker, in mm

SPECIES	SEALS (n)	MEAN LENGTH (mm)	SD (mm)
Harp seal	16	80.6	10.8
Harbor seal	15	86.5	10.5
Ringed seal	16	79.6	10.9
Spotted seal	9	91.2	7.9
Gray seal	8	73.8	12.5
…	…	…	…

One whisker per seal; only each species' n, mean and SD were published

Example: spotted seal whiskers

Estimate μ, the mean whisker length of all spotted seals, from the 9 that were measured.

n = 9; $\bar{x}$ = 91.2 mm; s = 7.9 mm

(a) z or t? How many degrees of freedom?
(b) What is the standard error?
(c) What is the 95% confidence interval for μ?

(a) t, because σ is unknown; df = 9 – 1 = 8, t = 2.306
(b) 7.9 / √9 ≈ 2.63 mm
(c) 91.2 ± 2.306(2.63) = 91.2 ± 6.06: 85.1 to 97.3 mm

Data: watermelon citrulline

Korea's national gene bank keeps seed of 100 watermelon varieties from around the world. Assefa et al. (2020) measured citrulline, the amino acid named after watermelon (Citrullus).

X = citrulline in a variety's fruit flesh, in mg per g of dry weight

VARIETY	ORIGIN	CITRULLINE (mg/g)
IT120008	Taiwan	27.16
IT199814	Kazakhstan	25.83
IT203017	Russia	26.38
IT203029	Kazakhstan	24.01
IT203627	unknown	32.79
…	…	…

A random sample of n = 12 of the 100 gene-bank varieties

Example: citrulline, 90% and 99%

Estimate μ, the mean citrulline of all 100 gene-bank varieties, from the sample of 12.

n = 12; $\bar{x}$ = 27.07 mg/g; s = 5.27 mg/g

(a) What are the degrees of freedom and the standard error?
(b) What are the 90% and 99% confidence intervals for μ?
(c) Which is wider, and why?

(a) df = 11; SE = 5.27 / √12 ≈ 1.52
(b) 90%: 27.07 ± 1.796(1.52): 24.34 to 29.80
99%: 27.07 ± 3.106(1.52): 22.35 to 31.79
(c) The 99% interval: more confidence needs a larger t

What you should know

confidence level

$C = 1 - \alpha$; $\alpha/2$ is left in each tail

critical value $z_{1-\alpha/2}$

area $1 - \alpha/2$ to its left: 1.645, 1.960, 2.576 for 90%, 95%, 99%

σ unknown

use s and t with df = n – 1: $\bar{x} \pm t\, s / \sqrt{n}$

width

wider for higher confidence; narrower for larger n